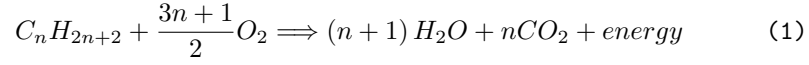


Bell Nozzle Rocket Engine Equation Database

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Propellant Stoichiometry of Hydrocarbon-Oxygen Propellant [1]

$$\dot{m} = \frac{m}{t} \quad (2)$$

Mass Flow Rate

$$\frac{1}{\gamma - 1} = \sum \frac{\text{mol}\%}{\gamma_i - 1} \quad (3)$$

Heat Capacity Ratio of a Multi-Compound Gas [2]

$$\frac{P}{P_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-\frac{\gamma}{\gamma - 1}} \quad (4)$$

Isentropic Pressure Relation [3]

$$\frac{T}{T_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-1} \quad (5)$$

Isentropic Temperature Relation [3]

$$\frac{\rho}{\rho_s} = \left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{-\frac{1}{\gamma - 1}} \quad (6)$$

Isentropic Density Relation [3]

$$P = \rho R_s T \quad (7)$$

Ideal Gas Law

$$Ma = \frac{v}{c} \quad (8)$$

Mach Velocity [3]

$$c = \sqrt{\gamma \frac{P}{\rho}} = \sqrt{\gamma R_s T} \quad (9)$$

Speed of Sound [3]

$$q = \frac{1}{2} \rho v^2 = \frac{\gamma}{2} P Ma^2 \quad (10)$$

Dynamic Pressure [3]

$$\epsilon = \left(\frac{\gamma + 1}{2}\right)^{\frac{\gamma + 1}{2(\gamma - 1)}} Ma_e \left(1 + \frac{\gamma - 1}{2} Ma_e^2\right)^{-\frac{\gamma + 1}{2(\gamma - 1)}} \quad (11)$$

Area Ratio [3]

$$P_c = \frac{\dot{m}\sqrt{T_t}}{A_t} \sqrt{\frac{R_s}{\gamma}} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad (12)$$

Chamber Pressure [4]

$$A_t = \frac{\dot{m}}{\rho_t v_t} \quad (13)$$

Throat Area [5]

$$0 = \frac{Ma_t}{Ma_e} \left(\frac{1 + \left(\frac{\gamma-1}{2} Ma_e^2 \right)}{1 + \left(\frac{\gamma-1}{2} Ma_t^2 \right)} \right)^{\frac{\gamma+1}{2(\gamma-1)}} - \epsilon \quad (14)$$

Isentropic Gas Relations [6]

References

- [1] J. Hawley, "The stoichiometry of burning hydrocarbon fuels." https://jimhawley.ca/downloads/Diesel/The_stoichiometry_of_burning_hydrocarbon_fuels.pdf, 2020.
- [2] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." <https://chemistry.stackexchange.com/a/166796>, 2022.
- [3] AMES Research Staff, "Equations, tables, and charts for compressible flow," *National Advisory Committee for Aeronautics*, 1953.
- [4] G. R. Center, "Rocket thrust summary." <https://www.grc.nasa.gov/WWW/k-12/airplane/rktthsum.html>, 2021.
- [5] B. Brady, "Cryo rocket." <https://www.cryo-rocket.com>, 2020.
- [6] O. B. Robert D. Zucker, *Fundamentals of Gas Dynamics*. Wiley, 2019.